

corrigé zakito

Énoncé

Autocorrection A

1

$$a_0 = S_0, \quad a_n = S_n - S_{n-1} \quad (n \geq 1).$$

2

$$(i) S_{n+1}; \quad (ii) S_{2n} - S_n; \quad (iii) 2S_n; \quad (iv) S_n - n - 1; \quad (v) S_n - \frac{n(n+1)}{2}.$$

Exercice 1

$$\begin{aligned} (i) \quad & \sum_{k=0}^n (-1)^k = \frac{1 - (-1)^{n+1}}{2}; \\ (ii) \quad & \sum_{k=2}^{n+1} k = \frac{(n+1)(n+2)}{2} - 1; \\ (iii) \quad & \sum_{k=1}^n 2k = 2 \frac{n(n+1)}{2} = n(n+1); \\ (iv) \quad & \sum_{k=n+1}^{2n} 2k = 2n(2n+1) - n(n+1) = n(3n+1); \\ (v) \quad & \sum_{k=0}^n 2^k = 2^{n+1} - 1; \\ (vi) \quad & \sum_{k=n+1}^{2n} 2^k = 2^{2n+1} - 2^{n+1}; \\ (vii) \quad & \sum_{k=0}^{2n} (-1)^k k = \sum_{j=1}^n (2j - (2j-1)) = n; \\ (viii) \quad & \sum_{k=0}^n (-1)^{n-k} = \sum_{j=0}^n (-1)^j = \frac{1 + (-1)^n}{2}. \end{aligned}$$

Exercice 2

$$\sum_{k=1}^n (x_k - \xi) = 0 \iff \sum_{k=1}^n x_k = n\xi \iff \xi = \frac{1}{n} \sum_{k=1}^n x_k.$$

Exercice 3

$$\begin{aligned} \sum_{p=1}^n \ln \left(1 + \frac{1}{p} \right) &= \sum_{p=1}^n (\ln(p+1) - \ln p) = \ln(n+1). \\ \sum_{k=0}^n k k! &= \sum_{k=0}^n ((k+1)! - k!) = (n+1)! - 1. \\ \sum_{k=2}^n \frac{1}{k^2 - 1} &= \frac{1}{2} \sum_{k=2}^n \left(\frac{1}{k-1} - \frac{1}{k+1} \right) = \frac{3}{4} - \frac{1}{2n} - \frac{1}{2(n+1)} \quad (n \geq 1). \end{aligned}$$

Pour $n = 0$, la somme vide vaut 0.

Exercice 4

$$T_j = \sum_{k=0}^n k^j, \quad (n+1)^4 = 4T_3 + 6T_2 + 4T_1 + n + 1.$$

$$T_1 = \frac{n(n+1)}{2}, \quad T_2 = \frac{n(n+1)(2n+1)}{6}.$$

$$4T_3 = (n+1)^4 - n(n+1)(2n+1) - 2n(n+1) - (n+1) = n^2(n+1)^2.$$

$$T_3 = \frac{n^2(n+1)^2}{4}.$$

Exercice 5

- (i) $(n^2 - n)^{2001}$;
- (ii) $2^n n!$;
- (iii) $(n!)^2$;
- (iv) $\left(\frac{(2n)!}{n!}\right)^2$;
- (v) $\frac{(2n+1)!}{2^n n!}$;
- (vi) $\exp\left(-\sum_{k=0}^n k\right) = e^{-n(n+1)/2}$;
- (vii) $2^{\sum_{k=0}^n k} = 2^{n(n+1)/2}$;
- (viii) $\prod_{k=2}^n \frac{k-1}{k} = \frac{1}{n} \quad (n \geq 1)$;
- (ix) $\prod_{k=2}^n \frac{k-1}{k} \frac{k+1}{k} = \frac{n+1}{2n} \quad (n \geq 1)$;
- (x) $\frac{(n+2)!}{2n!} = \frac{(n+1)(n+2)}{2}$;
- (xi) $\prod_{k=1}^n (2k-1)(2k+1) = \frac{(2n)!(2n+1)!}{2^{2n}(n!)^2}$;
- (xii) $\left(\prod_{k=1}^n k\right) \left(\prod_{k=1}^n (n+1-k)\right) = (n!)^2$.

Pour $n = 0$, les produits (viii) et (ix) valent 1.

Exercice 6

$$A = \prod_{i=1}^k u_i, \quad B = \prod_{j=k+1}^n u_j, \quad u_i \leq u_j \quad (i \leq k < j).$$

$$\prod_{i=1}^k \prod_{j=k+1}^n u_i \leq \prod_{i=1}^k \prod_{j=k+1}^n u_j \Rightarrow A^{n-k} \leq B^k.$$

$$A^n \leq A^k B^k = (AB)^k.$$

Exercice 7

$$\frac{a^n - 1}{a - 1} = \sum_{k=0}^{n-1} a^k \leq \sum_{k=0}^{n-1} a^{n-1} = na^{n-1}.$$

Exercice 8

1

$$S_{n+1} = S_n + (n+1)q^{n+1},$$

$$S_{n+1} = q \sum_{j=0}^n (j+1)q^j = qS_n + q \frac{1-q^{n+1}}{1-q}.$$

$$S_n = \frac{q - (n+1)q^{n+1} + nq^{n+2}}{(1-q)^2}.$$

2

$$(q-1)S_n = nq^{n+1} - \sum_{k=1}^n q^k = nq^{n+1} - \frac{q - q^{n+1}}{1-q}.$$

La division par $q-1$ donne la même expression.

3

$$\sigma(x) = \frac{1-x^{n+1}}{1-x}, \quad \sigma'(x) = \frac{1 - (n+1)x^n + nx^{n+1}}{(1-x)^2}.$$

$$S_n = q\sigma'(q) = \frac{q - (n+1)q^{n+1} + nq^{n+2}}{(1-q)^2}.$$

Autocorrection B

$$\sum_{k=0}^n (-1)^k \binom{n}{k} = (1-1)^n = \begin{cases} 1 & n=0, \\ 0 & n \geq 1, \end{cases}$$

$$\sum_{k=0}^n \frac{(-1)^k}{2^k} \binom{n}{k} = (1 - \frac{1}{2})^n = 2^{-n}, \quad \sum_{k=0}^n 2^{2k} \binom{n}{k} = (1+4)^n = 5^n.$$

Exercice 9

$$P(X) = \frac{X^5}{5} + \frac{X^4}{2} + \frac{X^3}{3} - \frac{X}{30}, \quad P(0) = 0,$$

$$P(n) - P(n-1) = n^4 \Rightarrow P(n) = \sum_{k=1}^n k^4 \in \mathbb{N}.$$

Exercice 10

$$E + O = \sum_{k=0}^n \binom{n}{k} = 2^n, \quad E - O = \sum_{k=0}^n (-1)^k \binom{n}{k} = 0.$$

$$E = O = 2^{n-1}.$$

Exercice 11

$$\binom{n}{k} \binom{n-k}{p-k} = \frac{n!}{k!(p-k)!(n-p)!} = \binom{n}{p} \binom{p}{k}.$$

$$\sum_{k=0}^p \binom{n}{k} \binom{n-k}{p-k} = 2^p \binom{n}{p}.$$

Il manque $\binom{n}{p}$ dans la première égalité imprimée : $n=2, p=1$ donne $4 \neq 2$.

$$p \geq 1 \Rightarrow \sum_{k=0}^p (-1)^k \binom{n}{k} \binom{n-k}{p-k} = \binom{n}{p} (1-1)^p = 0.$$

Exercice 12

$$A_n = \sum_{k \geq 0} \binom{n-k}{k}, \quad \binom{a}{b} = 0 \quad \text{si } (a, b) \in \mathbb{Z}^2 \text{ et } 0 \leq b \leq a \text{ est faux.}$$

$$A_0 = A_1 = 1, \quad A_n = \sum_{k \geq 0} \binom{n-1-k}{k} + \sum_{k \geq 1} \binom{n-1-k}{k-1} = A_{n-1} + A_{n-2} \quad (n \geq 2).$$

$$A_n = F_{n+1}.$$

Exercice 13

$$S_0 = S_1 = 1,$$

$$S_n = \sum_{k \geq 0} (-1)^k \binom{n-1-k}{k} + \sum_{k \geq 1} (-1)^k \binom{n-1-k}{k-1} = S_{n-1} - S_{n-2} \quad (n \geq 2).$$

$$S_{n+3} = -S_n, \quad S_{n+6} = S_n.$$

$$(S_{6m}, \dots, S_{6m+5}) = (1, 1, 0, -1, -1, 0).$$

Exercice 14

1

$$\binom{2p+2}{p+1} = \frac{2p+2}{p+1} \binom{2p+1}{p+1} = 2 \binom{2p+1}{p+1}.$$

2

$$\begin{aligned} S_{n+1} &= \sum_{k=0}^{n+1} \frac{\binom{n+k}{k} + \binom{n+k}{k-1}}{2^k} \\ &= S_n + \frac{\binom{2n+1}{n+1}}{2^{n+1}} + \frac{1}{2} \left(S_{n+1} - \frac{\binom{2n+2}{n+1}}{2^{n+1}} \right) = S_n + \frac{1}{2} S_{n+1}. \end{aligned}$$

$$S_{n+1} = 2S_n, \quad S_0 = 1 \Rightarrow S_n = 2^n.$$

Exercice 15

1

$$\frac{\binom{n}{k+1}}{\binom{n}{k}} = \frac{n-k}{k+1}.$$

$$\binom{n}{k+1} \begin{cases} > \binom{n}{k} & 2k < n-1, \\ = \binom{n}{k} & 2k = n-1, \\ < \binom{n}{k} & 2k > n-1. \end{cases}$$

2

$$2^{2n} = \sum_{k=0}^{2n} \binom{2n}{k} \leq (2n+1) \binom{2n}{n} \Rightarrow \binom{2n}{n} \geq \frac{2^{2n}}{2n+1}.$$

Exercice 16

$$m \geq 1 \Rightarrow \prod_{j=0}^{k-1} (m+j) = k! \binom{m+k-1}{k}.$$

Si un facteur est nul, le produit est nul. Si tous sont négatifs,

$$\prod_{j=0}^{k-1} (m+j) = (-1)^k \prod_{j=0}^{k-1} (-m-j),$$

un produit de k entiers positifs consécutifs au signe près.

Exercice 17

1

$$\frac{n}{k} \binom{n-1}{k-1} = \frac{n!}{k!(n-k)!} = \binom{n}{k}.$$

2

$$n \geq 1 : \quad \sum_{k=0}^n k \binom{n}{k} = n \sum_{j=0}^{n-1} \binom{n-1}{j} = n2^{n-1}.$$

La deuxième somme imprimée comporte $\binom{k}{n}$:

$$\sum_{k=0}^n k 2^k \binom{k}{n} = n2^n.$$

Avec $\binom{n}{k}$, la formule serait

$$\sum_{k=0}^n k 2^k \binom{n}{k} = 2n \sum_{j=0}^{n-1} 2^j \binom{n-1}{j} = 2n3^{n-1}.$$

Les deux sommes pondérées sont nulles pour $n = 0$.

$$\sum_{k=0}^n \frac{1}{k+1} \binom{n}{k} = \frac{1}{n+1} \sum_{j=1}^{n+1} \binom{n+1}{j} = \frac{2^{n+1} - 1}{n+1}.$$

Exercice 18

1

$$\sum_{k=0}^n \binom{n}{k} x^k = (1+x)^n \Rightarrow \sum_{k=1}^n k \binom{n}{k} x^{k-1} = n(1+x)^{n-1}.$$

$$x = 1 \Rightarrow \sum_{k=0}^n k \binom{n}{k} = n2^{n-1}.$$

2

$$n \geq 2 : \quad \sum_{k=0}^n k(k-1) \binom{n}{k} = n(n-1)2^{n-2}.$$

$$\sum_{k=0}^n k^2 \binom{n}{k} = n(n-1)2^{n-2} + n2^{n-1} = n(n+1)2^{n-2}.$$

Pour $n = 1$, la somme et la formule valent 1.

Exercise 19

1

$$\binom{k}{p} = \binom{k+1}{p+1} - \binom{k}{p+1} \Rightarrow \sum_{k=p}^n \binom{k}{p} = \binom{n+1}{p+1}.$$

2

$$\sum_{k=1}^n k(k+1)(k+2) = 6 \sum_{k=1}^n \binom{k+2}{3} = 6 \binom{n+3}{4} = \frac{n(n+1)(n+2)(n+3)}{4}.$$

$$k^3 = 6 \binom{k}{3} + 6 \binom{k}{2} + \binom{k}{1},$$

$$\sum_{k=0}^n k^3 = 6 \binom{n+1}{4} + 6 \binom{n+1}{3} + \binom{n+1}{2} = \frac{n^2(n+1)^2}{4}.$$

Exercise 20

$$A = \sum_{k=1}^n k = \frac{n(n+1)}{2}, \quad B = \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}.$$

- (i) $\sum_{j=1}^n j = A;$
- (ii) $nA = \frac{n^2(n+1)}{2};$
- (iii) $\sum_{j=1}^n \frac{j(j+1)}{2} = \frac{B+A}{2} = \frac{n(n+1)(n+2)}{6};$
- (iv) $\sum_{j=1}^n \left(\frac{j(j+1)}{2} + j^2 \right) = \frac{3B+A}{2} = \frac{n(n+1)^2}{2};$
- (v) $2nB + 2A^2 = \frac{n^2(n+1)(7n+5)}{6};$
- (vi) $\frac{1}{2}(A^2 + B) = \frac{n(n+1)(n+2)(3n+1)}{24};$
- (vii) $nA - nA = 0;$
- (viii) $\sum_{j=1}^n \left(j^2 - \frac{j(j+1)}{2} \right) = \frac{B-A}{2} = \frac{n(n^2-1)}{6};$
- (ix) $2 \sum_{1 \leq i \leq j \leq n} (j-i) = \frac{n(n^2-1)}{3};$
- (x) $\sum_{j=1}^n \left(\sum_{i=0}^j \binom{j}{i} - 1 \right) = \sum_{j=1}^n (2^j - 1) = 2^{n+1} - n - 2;$
- (xi) $\sum_{j=1}^n \frac{1}{j} \frac{j(j+1)}{2} = \frac{A+n}{2} = \frac{n(n+3)}{4};$
- (xii) $\sum_{j=1}^n \frac{1}{j} \frac{j(j+1)(2j+1)}{6} = \frac{2B+3A+n}{6} = \frac{n(4n^2+15n+17)}{36};$
- (xiii) $\sum_{k=1}^n k(2k-1) = 2B - A = \frac{n(n+1)(4n-1)}{6};$
- (xiv) $\sum_{k=1}^n k(2(n-k)+1) = (2n+1)A - 2B = \frac{n(n+1)(2n+1)}{6};$
- (xv) $\left(\sum_{i=1}^n 3^i \right)^2 = \frac{9(3^n-1)^2}{4}.$

Exercice 21

- (i) $\prod_{1 \leq i, j \leq n} x^{i+j} = x^{\sum_{i,j} (i+j)} = x^{n^2(n+1)} \quad (n \geq 1).$
- (ii) $\prod_{j=1}^n \prod_{i=1}^n i^j = \prod_{j=1}^n (n!)^j = (n!)^{n(n+1)/2}.$
- (iii) $\prod_{i,j} ij = \left(\prod_i i^n \right) \left(\prod_j j^n \right) = (n!)^{2n}.$

Pour $n = 0$, les trois produits vides valent 1.

Exercice 22

$$m \in \{1, \dots, n\} \Rightarrow m = \sum_{r=1}^n \mathbf{1}_{r \leq m}.$$

$$\min(i, \max(j, k)) \geq r \iff i \geq r \text{ et } (j \geq r \text{ ou } k \geq r).$$

$$\begin{aligned} \sum_{i,j,k=1}^n \min(i, \max(j, k)) &= \sum_{r=1}^n (n-r+1)(n^2 - (r-1)^2). \\ &= \sum_{s=0}^{n-1} (n-s)(n^2 - s^2) = \frac{n^3(n+1)}{2} - \frac{n^2(n-1)(2n-1)}{6} + \frac{n^2(n-1)^2}{4} \\ &= \frac{n^2(n+1)(5n+1)}{12}. \end{aligned}$$

Exercice 23

1

$$\begin{aligned} \sum_{0 \leq i < j \leq n} i^p &= \sum_{i=0}^n (n-i)i^p = nS_n^{(p)} - S_n^{(p+1)}, \\ \sum_{0 \leq i < j \leq n} i^p &= \sum_{j=1}^n \sum_{i=0}^{j-1} i^p = \sum_{k=0}^{n-1} S_k^{(p)}. \end{aligned}$$

2

$$\begin{aligned} nS_n^{(3)} - S_n^{(4)} &= \frac{1}{4} \sum_{k=0}^{n-1} (k^4 + 2k^3 + k^2). \\ 4nS_n^{(3)} - 4S_n^{(4)} &= S_n^{(4)} - n^4 + 2S_n^{(3)} - 2n^3 + S_n^{(2)} - n^2. \\ 5S_n^{(4)} &= (4n-2) \frac{n^2(n+1)^2}{4} + n^4 + 2n^3 + n^2 - \frac{n(n+1)(2n+1)}{6}. \\ S_n^{(4)} &= \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}. \end{aligned}$$

Exercice 24

1

$$[1, n+1]^2 = [1, n]^2 \sqcup ([1, n] \times \{n+1\}) \sqcup (\{n+1\} \times [1, n]) \sqcup \{(n+1, n+1)\},$$

où les intervalles désignent des intervalles d'entiers. La somme sur cette partition donne l'égalité demandée.

2

$$\begin{aligned} S_n &= \left(\sum_{k=1}^n k \right)^2, \\ S_{n+1} - S_n &= 2(n+1) \sum_{k=1}^n k + (n+1)^2 = n(n+1)^2 + (n+1)^2 = (n+1)^3. \\ S_0 = 0 &\Rightarrow S_n = \sum_{k=1}^n k^3 = \left(\sum_{k=1}^n k \right)^2. \end{aligned}$$

Exercice 25

$$\left(\sum_i x_i \right) \left(\sum_j x_j^{-1} \right) - n^2 = \sum_{i < j} \left(\frac{x_i}{x_j} + \frac{x_j}{x_i} - 2 \right) = \sum_{i < j} \frac{(x_i - x_j)^2}{x_i x_j} \geq 0.$$

Égalité si et seulement si $x_1 = \dots = x_n$.

Exercice 26

$$2n \sum_k a_k b_k - 2 \left(\sum_i a_i \right) \left(\sum_j b_j \right) = \sum_{i,j} (a_i - a_j)(b_i - b_j) \geq 0.$$

$$\frac{1}{n} \sum_k a_k b_k \geq \left(\frac{1}{n} \sum_i a_i \right) \left(\frac{1}{n} \sum_j b_j \right) \quad (n \geq 1).$$

Exercice 27

$$S(n+1) - S(n) = a_{n+1} \geq 0.$$

$$S \text{ injective} \iff \forall n \geq 1, a_n \geq 1.$$

$$S \text{ surjective} \Rightarrow a_0 = S(0) = 0, \quad a_n \in \{0, 1\} \quad (n \geq 1), \quad S(n) \rightarrow +\infty.$$

Réciproquement, ces conditions interdisent tout entier manquant entre deux valeurs consécutives de S , et S est non bornée.

$$S \text{ surjective} \iff a_0 = 0, \quad \forall n \geq 1, a_n \in \{0, 1\}, \quad \#\{n \geq 1 : a_n = 1\} = +\infty.$$

$$S \text{ bijective} \iff a_0 = 0 \text{ et } \forall n \geq 1, a_n = 1.$$

Exercice 28

1(a)

$$\psi(A) = 1 \Rightarrow w_0 = 1.$$

$$T_k = \sum_{j=0}^k w_j, \quad w_{k+1} > T_k + 1 \Rightarrow T_k + 1 \notin \psi(\mathcal{P}_f(\mathbb{N})).$$

En effet, une somme utilisant un indice $> k$ dépasse $T_k + 1$; les autres sont au plus T_k .

1(b)

$$\{0, 1\} \subset \psi(\mathcal{P}(\{0\})).$$

$$[0, T_k] \cap \mathbb{N} \subset \psi(\mathcal{P}(\{0, \dots, k\}))$$

$$\Rightarrow ([0, T_k] \cup [w_{k+1}, w_{k+1} + T_k]) \cap \mathbb{N} \subset \psi(\mathcal{P}(\{0, \dots, k+1\})).$$

$$w_{k+1} \leq T_k + 1 \Rightarrow [0, T_{k+1}] \cap \mathbb{N} \subset \psi(\mathcal{P}(\{0, \dots, k+1\})).$$

$$T_k \geq k+1 \rightarrow +\infty \Rightarrow \psi \text{ surjective.}$$

2

$$w_{k+1} \leq T_k \Rightarrow \exists A \subset \{0, \dots, k\}, \quad \psi(A) = w_{k+1} = \psi(\{k+1\}),$$

ce qui contredit l'injectivité. Ainsi

$$w_{k+1} = T_k + 1, \quad w_0 = 1 \Rightarrow w_k = 2^k.$$

Réciproquement, si $A \neq B$, soit $m = \max(A \Delta B)$; quitte à échanger A, B , $m \in A \setminus B$.

$$\psi(A) - \psi(B) \geq 2^m - \sum_{j=0}^{m-1} 2^j = 1 > 0.$$

$$\psi \text{ bijective} \iff \forall k, w_k = 2^k.$$